RevTrace: Coordination Algorithm Draft

Standalone source-grounded description, 31 October 2026

Thesis. RevTrace preserves valid service progress across trusted feedback revisions by independently validating complete call inputs, retaining original selection authority, and checking support for the Consumer's whole view.

Objects. For session s, $C_s = (F_s, P_s)$ is the Owner's installed selected cut. A frozen call j has input $q_j = (s, e_j, a_j, P_j^-, b_j, \chi_j)$, where a_j and b_j are its start and target frontiers and χ_j binds the immutable source/audio, target/EOS, model, decoding, and runtime. Its original UID, birth root, and producer $(slot, incarnation)$ never change on confirmation. J holds live owned calls and B holds strictly accepted complete original results. Typed ASR identity projections have a separate type; they are provisional dependency nodes, not complete original selection results. $W_s[e, f]$ records an actual selected cut at frontier f in session epoch e.

Input status, in priority order. Write start(j) for the complete closed start cut, including its frontier. The remaining inputs must retain their frozen bindings.

$$E(j) = \begin{cases} \textsc{Fenced} & e_j \neq e_s, \\ \textsc{Confirmed} & \text{start}(j) = C_s, \\ \textsc{Refuted} & W_s[e_s, a_j] \text{ exists and } W_s[e_s, a_j] \neq \text{start}(j), \\ \textsc{Covered} & b_j \leq F_s, \\ \textsc{Unknown} & \text{otherwise.} \end{cases}$$

Loss of an earlier ancestor's support does not refute a future complete input. Unknown grants neither commit authority nor a refutation-based cancellation proof. These labels explain the existing guards; they need not be stored as runtime states.

Whole-view support. $S(C, h)$ requires a complete, contiguous published dependency path from the new cut C to the actual whole view h, with exact state and frontier matches under one contract. A view may lose this support while a future call's input remains Unknown.

Algorithm 1: Existing full-call coordinator, with separate application

1. **Dispatch:** on a ready, unleased worker, prefer missing work from C_s; otherwise use an unduplicated legal conditional tail within the declared ahead bound. Require accepted audio through b_j. Freeze q_j and the original identities before sending the call.
2. **Receive:** bind the physical channel, incarnation, registered UID, session, phase, and semantic fingerprint. Locally reject foreign, fenced, retired, or exact duplicate messages; contradictory currently owned contents freeze writes. No local drop clears another lease.
3. For a valid typed ASR projection, append it only if its start is the current conditional tail. Publish a whole-view snapshot; do not record an original selected transition.
4. For a valid complete original result, verify the frozen start, target, original C/U record and output binding. Deny selection if its input is refuted, covered, or fenced; select immediately if confirmed, otherwise buffer it in B. For a buffered result, replace its typed prediction with the actual full output, retaining only a suffix that consumes that output contiguously; publish the resulting view. A validated terminal retires exactly its physical job.
5. **Select ready:** for an immediately confirmed result, or while B contains a result with $E(j) = \textsc{Confirmed}$, recheck the current parent and complete original record. Synchronize the bound parent/result/identity as a new durable fact. An uncertain outcome freezes writes and requires recovery.
6. Only after successful synchronization, install its result as C_s, record $W_s[e_s, F_s]$, reconcile the conditional chain by exact boundaries, and publish the fact plus the view snapshot for *this* installed root. Reclassify buffered results; keep unresolved results unselected.
7. **Actual Consumer:** verify Owner/publication order and the durable fact's identity, sequence and actual parent. Apply its certified cut; retain the existing ahead whole view only when $S(C_s, h)$ holds. Private publications require the current root and a complete path, and cannot replace a farther view with a shorter private snapshot.
8. Record the actual application receipt. Release its outbox references, while retaining references still used by the visible view. Input debt and original offer clocks survive cancellation, failure, and replay.

Worker loss and recovery. After confirmed OS exit, joined result and control readers, and old CUDA release, strictly admit complete receipts from the sealed old channel *before* fencing its incarnation. Then retire unfinished work and launch a higher incarnation. Already accepted complete buffers and supported views keep their original

identities; producer death does not make them new late messages. Owner recovery fences old ownership, advances the Owner namespace, verifies the durable prefix, and replays existing facts. Replay adds no inference or new selected edge.

Table 1: Separate decisions made by the existing coordinator.

Condition	Call / selected-state decision	View / lifecycle decision
Complete same-frontier input matches	Continue owned call; completed result may enter ordered selection with original UID/birth/incarnation	Check whole-view support separately; predicted end need not match original C/U
Actual same-frontier full input differs	Deny append/install; safe-point exit needs this bound refutation proof	Withdraw unsupported view; do not generalize to other future inputs
Future input unresolved, including earlier ancestor loss	Buffer or continue; no commit and no refutation-based cancellation	Withdraw a view if unsupported; input may later converge
Target covered or session fenced	Deny selection; retire only the exact matching old job	Keep accepted-input debt and original clocks
Complete original C/U and confirmed input	Durable sync, then Owner installation, then publication	Actual Consumer verifies and applies; snapshot belongs to each particular installed root
Producer exits	Seal/strictly drain before fencing; preserve accepted buffer identity	Retire unfinished work; replace after physical release; supported view can survive
Foreign/fenced/duplicate or owned contradiction	Local reject the former; freeze writes for the latter	Drop cannot release a newer lease; a valid old-session terminal retires only its own old lease
Owner restart / Consumer receipt	Replay checked existing facts without selecting again	ACK frees outbox references; visible dependencies stay pinned; replay is not a new result

Four distinct events. Let D be durable selection, C Owner installation, V actual Consumer application, and L physical release. D precedes privileged installation/application; neither C nor V selects the fact again. Logical invalidation and cancellation requests do not imply L. The core batch runtime requires all terminal rows plus device synchronization for `BatchDrained`. Native4f instead clears one ordinary busy job on its strictly matched complete/abandoned terminal; worker replacement additionally requires OS/channel/CUDA release evidence. These implementations share the distinction, not an identical release event.

Proposition 1 (Selected-prefix preservation, conditional)**.** *Assume sufficient declared cross-call state; faithful execution of frozen accepted inputs by the protected original program $\mathcal{A}$; exclusive serial selection with correctly bound current-session receipts; immutable audio/model/decoding bindings; unambiguous encoding and binding digests; storage honoring successful synchronization; and recovery of only a verified contiguous fact prefix. If every new durable fact binds a complete original result whose full input equals the preceding selected cut, and synchronization precedes installation and certified application, then the distinct durable selected facts form a trace permitted by $\mathcal{A}$. Replaying a fact adds no transition.*

Proof. The genesis cut is an initial state. Each appended fact is a bound complete original call on the actual preceding cut and accepted input, hence an allowed next edge. Speculation, projections, waiting, cancellation, rejection, collection and repeated delivery add no selected edge. Verified replay preserves the same fact identities. Induction yields the prefix. □

The relation of $\mathcal{A}$ may be nondeterministic. Equality with a fixed neural reference is separate measured evidence; arbitrary independent GPU reruns are not assumed bitwise identical. Provisional UI behavior is outside this projection. Its separate sufficient property is per application: an ahead whole view satisfying $S(C, h)$ is retained unchanged by that compatible Canonical application when no independent withdrawal mode is invoked. Neither property certifies human transcription accuracy, global text monotonicity, fault-time deadlines, or all Python/CUDA/OS behavior.

Source boundary. The algorithm describes unchanged native source `4f4c3cab`, particularly `fullspec_policy`, `resilience_policy`, `resilience_owner`, `resilience_consumer`, `resilience_storage`, and `resilience_channel_close`. Controller `a13fa0a3` only adds paired/resident orchestration. Buffer fixture `4eb03d6f` controls interleaving; reader `b8670073` separately rechecks unchanged raw evidence with zero new neural calls. Its mechanism result does not upgrade the original failed CLI or compound gates.